A Generalized Non-Equilibrium Bounce-Back Condition for Multi-Component DdQn Lattice Boltzmann Models with Wetting Applications.

D. Schrijver

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Abstract will come here...

I. INTRODUCTION

Introduction will come here...

II. KNUDSEN EXPANSION

In diffusive scaling, we get the following for a Knudsen expansion,

$$\begin{aligned}
f_i^{(0)} &= f_i^{\text{eq},(0)} = w_i \rho^{(0)}, \\
f_i^{(1)} &= f_i^{\text{eq},(1)} = w_i \rho^{(1)} + w_i c_{i\alpha} \rho^{(0)} \frac{u_\alpha^{(1)}}{c_s^2}, \\
f_i^{(2)} &= f_i^{\text{eq},(2)} - \frac{\tau}{Sh} c_{i\alpha} \partial_\alpha f_i^{\text{eq},(1)}, \\
f_i^{(3)} &= f_i^{\text{eq},(3)} - \tau \partial_t f_i^{\text{eq},(1)} - \frac{\tau}{Sh} c_{i\alpha} \partial_\alpha f_i^{\text{eq},(2)} \\
&\quad + \frac{\tau}{Sh^2} c_{i\alpha} \partial_\alpha \left(\left(\tau - \frac{\Delta t}{2} \right) c_{i\beta} \partial_\beta f_i^{\text{eq},(1)} \right)
\end{aligned} \tag{1}$$

where repeated Greek indices are implicitly summed over. At the boundary, the outgoing and parallel populations are governed by the LBE, which means that they are given by the equation above. The incoming populations are given by the NEBB,

$$\begin{aligned}
f_i^{\text{neq}} &= f_q^{\text{neq}} - N_i, \\
f_i &= f_q + 2 w_i \rho \frac{\boldsymbol{u} \cdot \boldsymbol{c}_i}{c_s^2} - N_i, \\
N_i &= \frac{2 w_i c_{i\alpha}}{c_s^2 A_\alpha} \left[\sum_j^{\text{par}} c_{j\alpha} f_j + (A_\alpha - 1) \rho u_\alpha \right],
\end{aligned} \tag{2}$$

where $A_\alpha = \frac{2}{c_s^2} \sum_j^{\text{in}} w_j c_{j\alpha}^2 = \delta_{\alpha n} + \frac{1}{3}(1 - \delta_{\alpha n})$. We give the moments of N_i,

$$\begin{aligned}
\sum_i^{\text{in}} N_i &= 0, \\
\sum_i^{\text{in}} c_{i\alpha} N_i &= \sum_i^{\text{par}} c_{i\alpha} f_i + (A_\alpha - 1) \rho u_\alpha
\end{aligned} \tag{3}$$

Rewritten bulk:

$$\begin{aligned}
f_i^{\text{neq},(0)} &= 0, \\
f_i^{\text{neq},(1)} &= 0, \\
f_i^{\text{neq},(2)} &= -\frac{\tau}{Sh} c_{i\alpha} \partial_\alpha f_i^{\text{eq},(1)}, \\
f_i^{\text{neq},(3)} &= -\tau \partial_t f_i^{\text{eq},(1)} - \frac{\tau}{Sh} c_{i\alpha} \partial_\alpha f_i^{\text{eq},(2)} \\
&\quad - \frac{\tau}{Sh} c_{i\alpha} \partial_\alpha \left(\left(1 - \frac{\Delta t}{2\tau} \right) f_i^{\text{neq},(2)} \right)
\end{aligned} \tag{4}$$

Rewritten boundary:

$$\begin{aligned}
f_i^{\text{neq},(0)} &= -N_i^{(0)} \\
f_i^{\text{neq},(1)} &= -N_i^{(1)}, \\
f_i^{\text{neq},(2)} &= \frac{\tau}{Sh} c_{i\alpha} \partial_\alpha f_q^{\text{eq},(1)} - N_i^{(2)}, \\
f_i^{\text{neq},(3)} &= -\tau \partial_t f_q^{\text{eq},(1)} + \frac{\tau}{Sh} c_{i\alpha} \partial_\alpha f_q^{\text{eq},(2)} \\
&\quad + \frac{\tau}{Sh} c_{i\alpha} \partial_\alpha \left(\left(1 - \frac{\Delta t}{2\tau} \right) f_q^{\text{neq},(2)} \right) - N_i^{(3)}
\end{aligned} \tag{5}$$

The bounced-back equilibrium is given as follows,

$$\begin{aligned}
f_q^{\text{eq},(1)} &= f_i^{\text{eq},(1)} - 2 w_i c_{i\alpha} \rho^{(0)} \frac{u_\alpha^{(1)}}{c_s^2} \\
f_q^{\text{eq},(2)} &= f_i^{\text{eq},(2)} - 2 w_i c_{i\alpha} \left(\rho^{(0)} \frac{u_\alpha^{(2)}}{c_s^2} + \rho^{(1)} \frac{u_\alpha^{(1)}}{c_s^2} \right)
\end{aligned} \tag{6}$$

The bounced-back negative equilibrium is given as follows,

$$\begin{aligned}
-f_q^{\text{eq},(1)} &= f_i^{\text{eq},(1)} - 2 w_i \rho^{(1)} \\
-f_q^{\text{eq},(2)} &= f_i^{\text{eq},(2)} - 2 w_i \left(\rho^{(2)} + \frac{\left(u_\alpha^{(1)} c_{i\alpha} \right)^2}{4 c_s^2} - \frac{u_\alpha^{(1)} u_\alpha^{(1)}}{2 c_s^2} \right)
\end{aligned} \tag{7}$$

Using the following equations along with the previous,

$$\begin{aligned}
\sum_i^{\text{in}} c_{i\alpha} N_i^{(0)} &= 0 \\
\sum_i^{\text{in}} c_{i\alpha} N_i^{(1)} &= 0 \\
\sum_i^{\text{in}} c_{i\alpha} N_i^{(2)} &= \sum_i^{\text{par}} c_{i\alpha} f_i^{\text{neq},(2)} \\
\sum_i^{\text{in}} c_{i\alpha} N_i^{(3)} &= \sum_i^{\text{par}} c_{i\alpha} f_i^{\text{neq},(3)}
\end{aligned} \tag{8}$$

Spurious effect for zeroth moment of order 2 in Knudsen,

$$-2\frac{\tau}{Sh}\sum_i^{\text{in}} w_i c_{i\alpha}\partial_\alpha \rho^{(1)} = -\frac{\tau}{Sh}\partial_n\left(c_s^2\rho^{(1)}\right) \tag{9}$$

Spurious effect for zeroth moment of order 3 in Knudsen,

$$\boldsymbol{\nabla} = \tau\rho^{(0)}\partial_t u_n^{(1)} \tag{10}$$

Mass loss:

$$\begin{aligned}
\Delta m(t+\Delta t) &= \sum_i^{\text{out}}\Big(f_i^*(t) - f_i(t+\Delta t)\Big) \\
f_i^*(\boldsymbol{x},t) &= f_i(\boldsymbol{x}+\boldsymbol{c}_i\Delta x, t+\Delta t) \\
&\approx f_i + \frac{\Delta t}{Sh} c_{i\alpha}\partial_\alpha f_i + \frac{\Delta t^2}{2Sh^2} c_{i\alpha} c_{i\beta}\partial_\alpha\partial_\beta f_i \\
\Delta m^{(2)} &= \frac{\Delta t}{Sh}\sum_i^{\text{out}} c_{i\alpha}\partial_\alpha f_i^{\text{eq},(1)} \\
&= \frac{\Delta t}{Sh}\Bigg(-\frac{1}{2}\partial_n\left(c_s^2\rho^{(1)}\right) + \frac{1}{2}\rho^{(0)}\partial_n u_n^{(1)} \\
&\qquad + \frac{1}{6}\rho^{(0)}(1-\delta_{\alpha n})\partial_\alpha u_\alpha^{(1)}\Bigg)
\end{aligned} \tag{11}$$